REACTIVE POWER COMPENSATION IN GRID CONNECTED PV CELL

EEE-306

POWER SYSTEM LABORATORY

GROUP:03

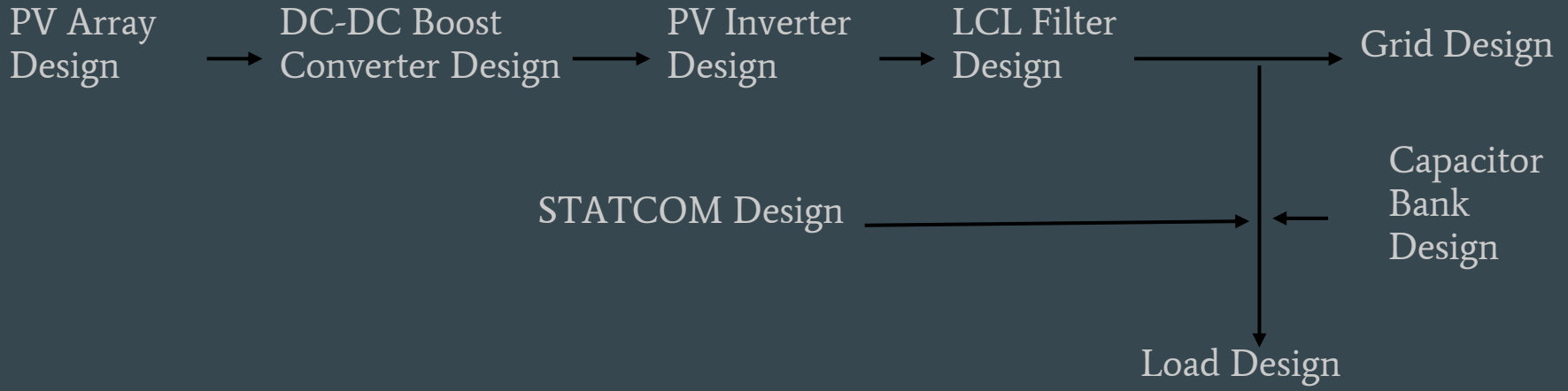
STUDENT ID: 2006157,2006158,

2006159,2006160, 2006161,2006163.

Objective

Reactive power compensation for a PV integrated grid system, using a fixed capacitor bank and Static Synchronous Compensator (STATCOM)

Roadmap


PV Array Design
DC-DC Boost Converter Design
PV Inverter Design
LCL Filter Design
Grid Design
Capacitor Bank Design
STATCOM Design
Load Design

PV Array

RATINGS:

100.345 kW, the open circuit voltage 363 V, short circuit current 368.48 A, maximum power point current is 345.45 A, and maximum power point voltage is 290 v.

PV array (mask) (link)

Implements a PV array built of strings of PV modules connected in parallel. Each string consists of modules connected in series.
Allows modeling of a variety of preset PV modules available from NREL System Advisor Model (Jan. 2014) as well as user-defined PV module.

Input 1 = Sun irradiance, in W/m2, and input 2 = Cell temperature, in deg.C.

Parameters | Advanced

Array data

Parallel strings: 47

Series-connected modules per string: 10

Display I-V and P-V characteristics of ...

array @ 25 deg.C & specified irradiances

Irradiances (W/m2): [1000 500 100]

Plot

Module data

Module: Solartech Renewables STR215

Maximum Power (W): 214.963

Cells per module (Ncell): 60

Open circuit voltage Voc (V): 36

Short-circuit current Isc (A): 7.73

Voltage at maximum power point Vmp (V): 28.7

Current at maximum power point Imp (A): 7.49

Temperature coefficient of Voc (%/deg.C): -0.41

Temperature coefficient of Isc (%/deg.C): 0.05

Model parameters

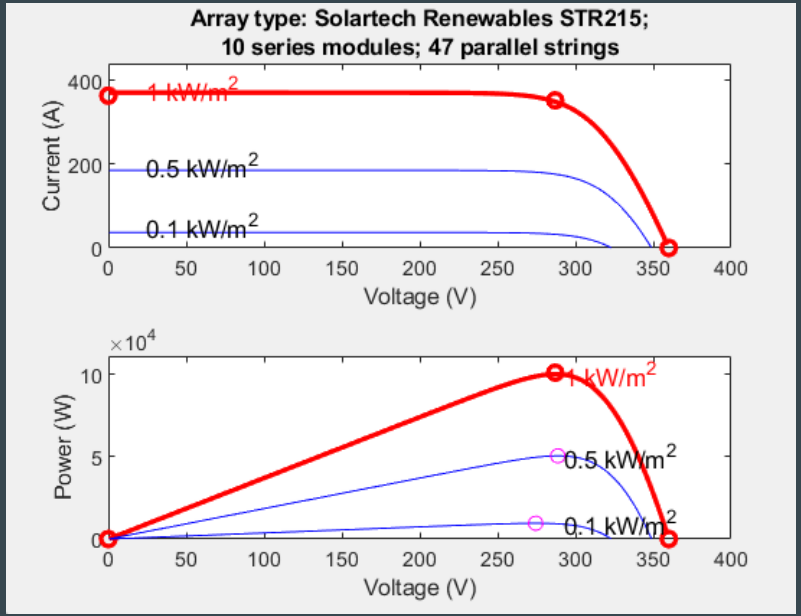
Light-generated current IL (A): 7.8848

Diode saturation current I0 (A): 1.3153e-09

Diode ideality factor: 1.0373

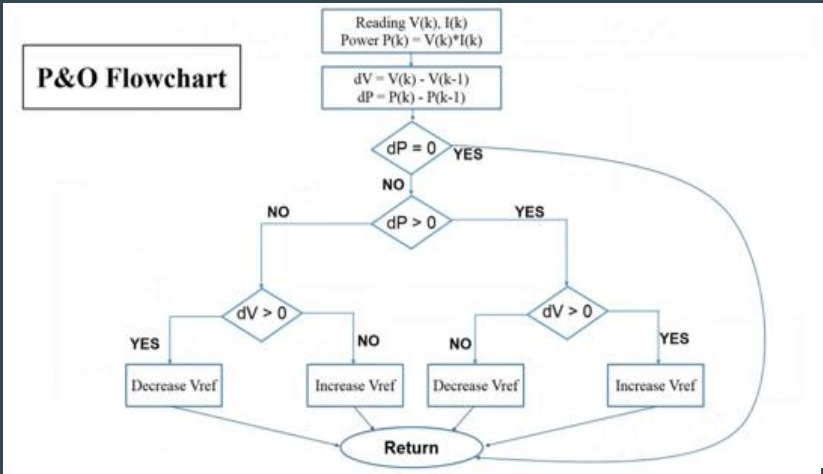
Shunt resistance Rsh (ohms): 5284.6435

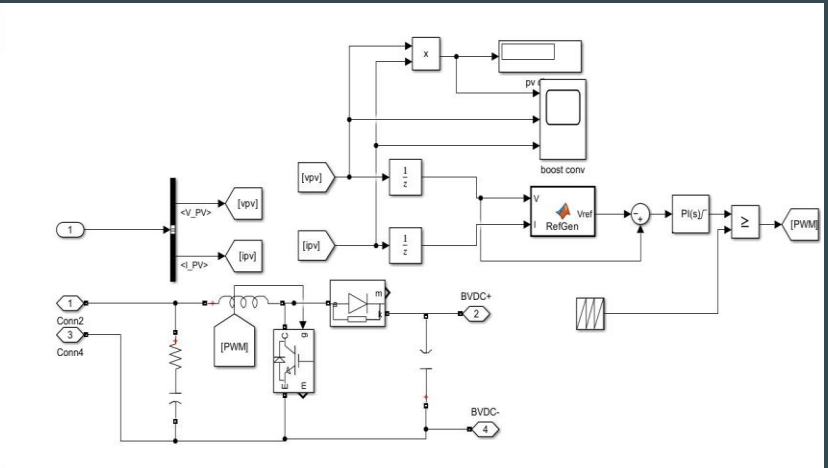
Series resistance Rs (ohms): 0.36886



DC-DC Boost Converter

- IGBT used as switch
- Switch Closed: I increases, magnetic field increases and Capacitor consumes energy
- Switch Opened: I decreases, polarity of magnetic field changes and capacitor releases energy
- Continuous cycles increases the voltage across Capacitor
- Diode: blocks back direction current flow
- IGBT is controlled with the help of PI controller and MPPT





The following equation used to design the inductor and capacitor of the direct current – direct current step-up converter in the solar PV system for grid integration

$$I_{outmax} = \frac{P}{V_{out}}$$

$$\Delta I_L = 0.07 \times I_{outmax} \times \frac{V_{out}}{V_{in}}$$

$$\Delta V_{out} = 0.007 \times V_{out}$$

$$L = \frac{V_{in} \times (V_{out} - V_{in})}{\Delta I_L \times F_s \times V_{out}}$$

$$C = \frac{I_{outmax} \times (1 - \frac{V_{in}}{V_{out}})}{\Delta V_{out} \times F_s}$$

L = filter inductor of dc-dc step-up converter,
C = filter capacitor of dc-dc step-up converter,
P=standard test condition power rating of solar PV &
Vin = PV voltage at standard test conditions,
$Ioutmax$ = maximum current of the converter at the terminal of dc-dc step-up converter,$Vout$= the output voltage of dc-dc step-up converter, ΔIL is the ripple inductor current of dc-dc step-up converter, $\Delta Vout$ is the ripple output voltage of dc dc and Fs= the switching frequency of dc-dc step-up converter.

The maximum current of the direct current – direct current step-up converter at the output terminal is,

$$I_{outmax} = \frac{100.345 \times 1000}{700} = 143.35\ A$$

The ripple inductor current of the direct current – direct current step-up converter is,

$$\Delta I_L = 0.07 \times 143.34 \times \frac{700}{290} = 24.21\ A$$

The ripple inductor voltage of the direct current – direct current step-up converter is,

$$\Delta V_{out} = 0.007 \times 700 = 4.9\ V$$

The designed value of the inductor of the direct current – direct current step-up converter is,

$$L = \frac{290 \times (700 - 290)}{24.21 \times 5000 \times 700} = 1.4\ mH$$

The designed value of the capacitor of the direct current – direct current step-up converter is,

$$C = \frac{143.35 \times (1 - \frac{290}{700})}{4.9 \times 5000} = 3400\ \mu F$$

```
function Vref = RefGen(V,I)


Vrefmax=363;
Vrefmin=0.0;
Vrefinit=300;
deltaVref = 1;
persistent Vold Pold Vrefold;

dataType = 'double';

if isempty (Vold)
    Vold=0;
    Pold=0;
    Vrefold=Vrefinit;
end
P=V*I;
dV=V - Vold;
dP= P - Pold;

if dP ~= 0
    if dP <0
        if dV <0
             Vref =Vrefold + deltaVref;
        else
            Vref= Vrefold - deltaVref;
        end
    else
        if dV <0
            Vref= Vrefold - deltaVref;
        else
            Vref =Vrefold + deltaVref;
        end
    end
else Vref= Vrefold;
end

if Vref>=Vrefmax | Vref<=Vrefmin

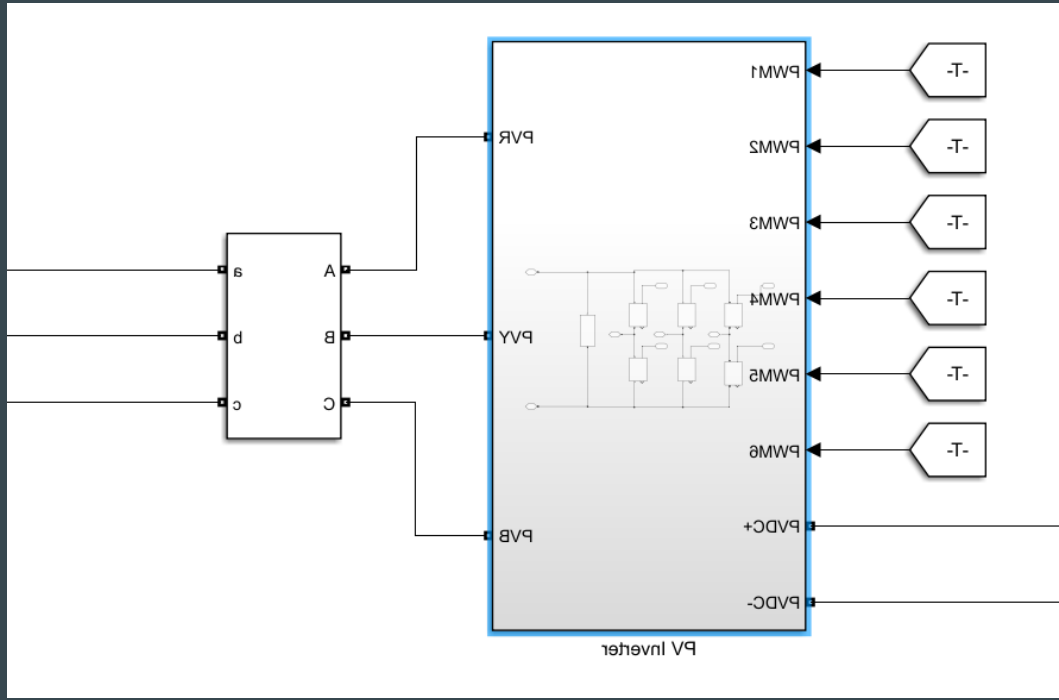
    Vref=Vrefold;
end

Vrefold=Vref;
Vold=V;
Pold=P;
```

PV Inverter

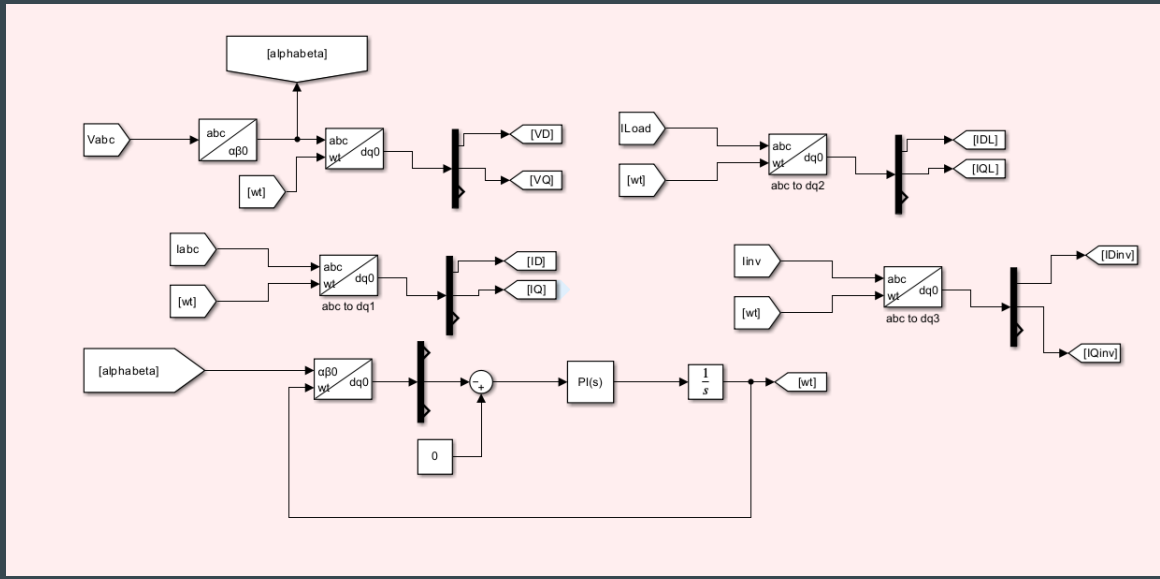
PV inverter is used for DC to AC voltage transformation.

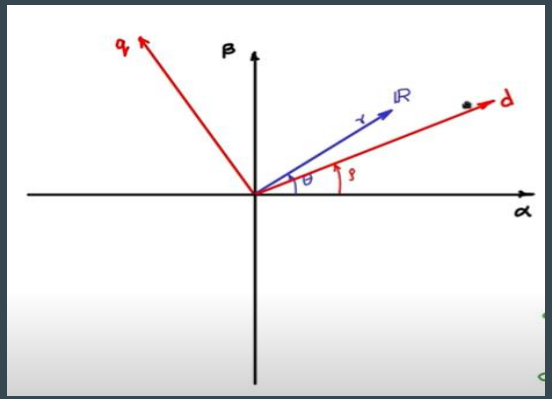
We used Sinusoidal PWM

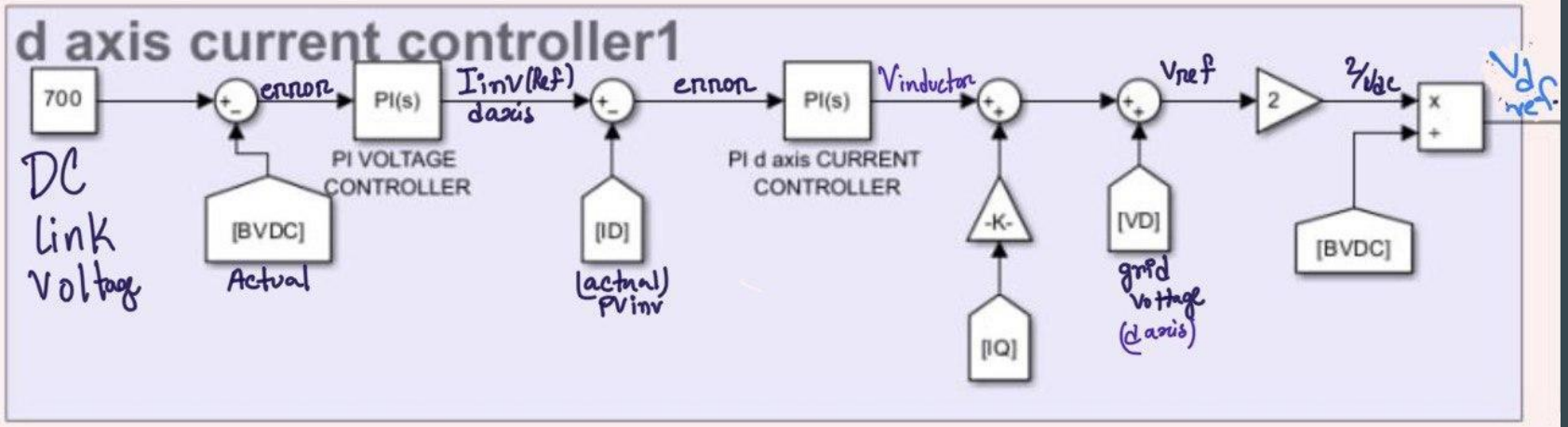


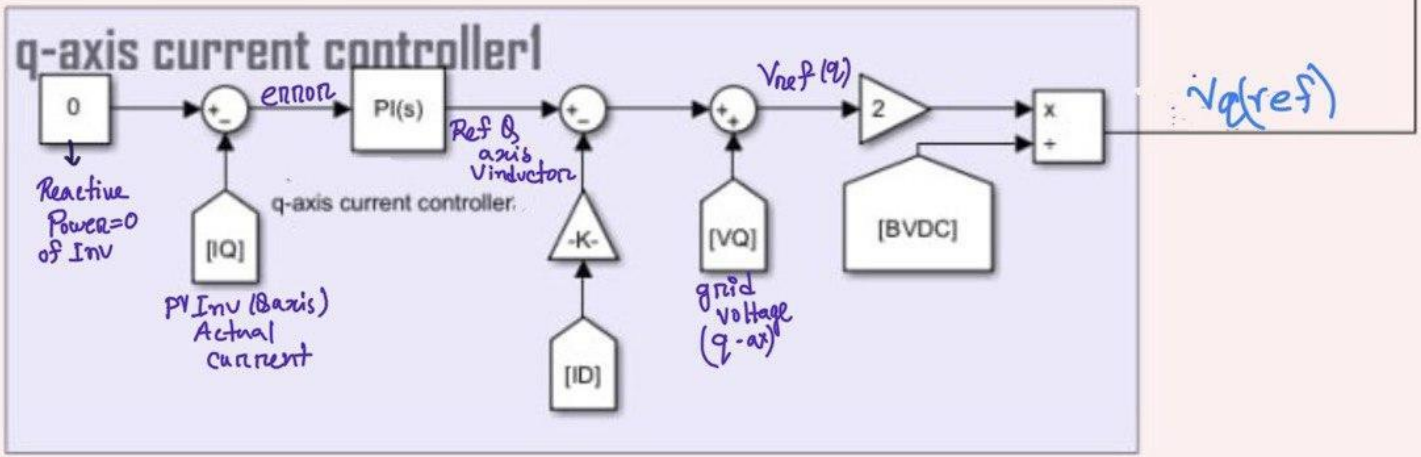
PV Inverter controller block:

- DC to AC conversion
- Clarke & Park trans.
- PLL

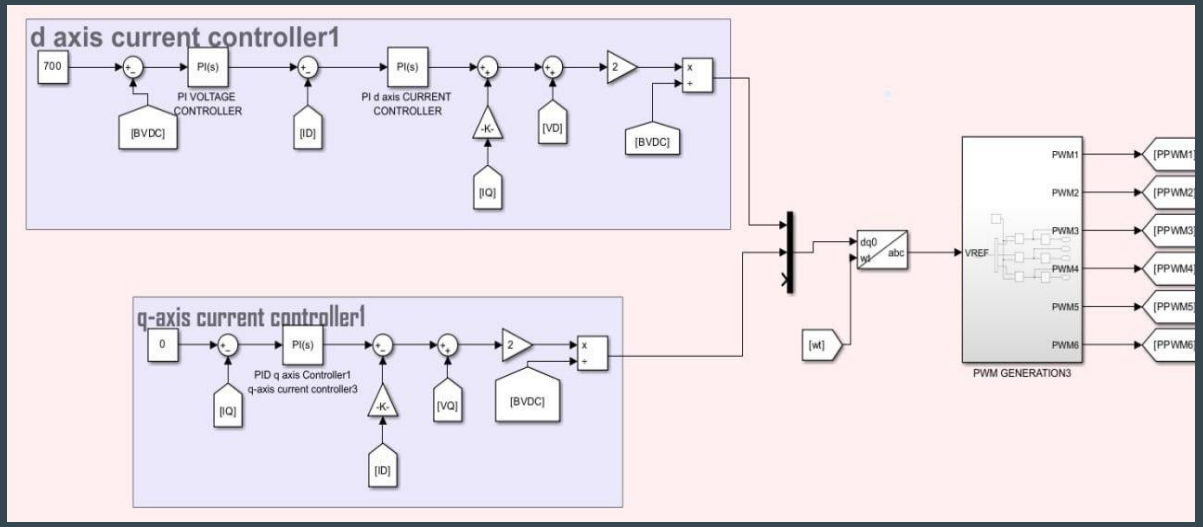


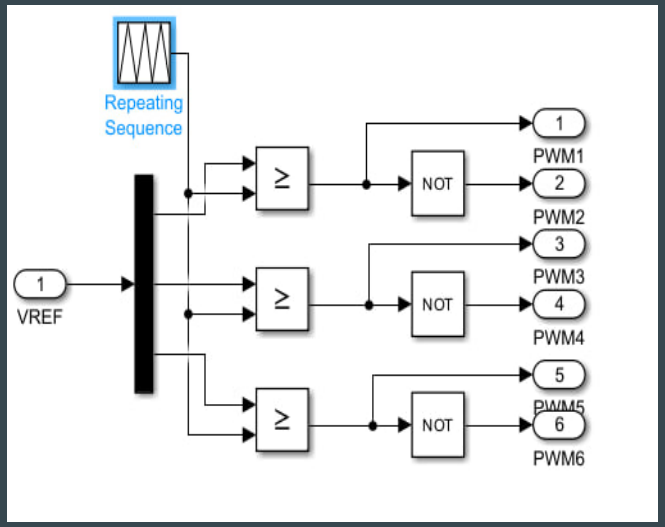



d axis current controller1
700
DC link Voltage
error
PI(s)
PI VOLTAGE CONTROLLER
[BVDC]
Actual
Iinv(Ref) daxis
[ID]
(actual) PVinv
error
PI(s)
PI d axis CURRENT CONTROLLER
Vinductor
-K-
[IQ]
Vref
[VD]
grid Voltage (d axis)
2
2/Vdc
[BVDC]
x
Vd ref

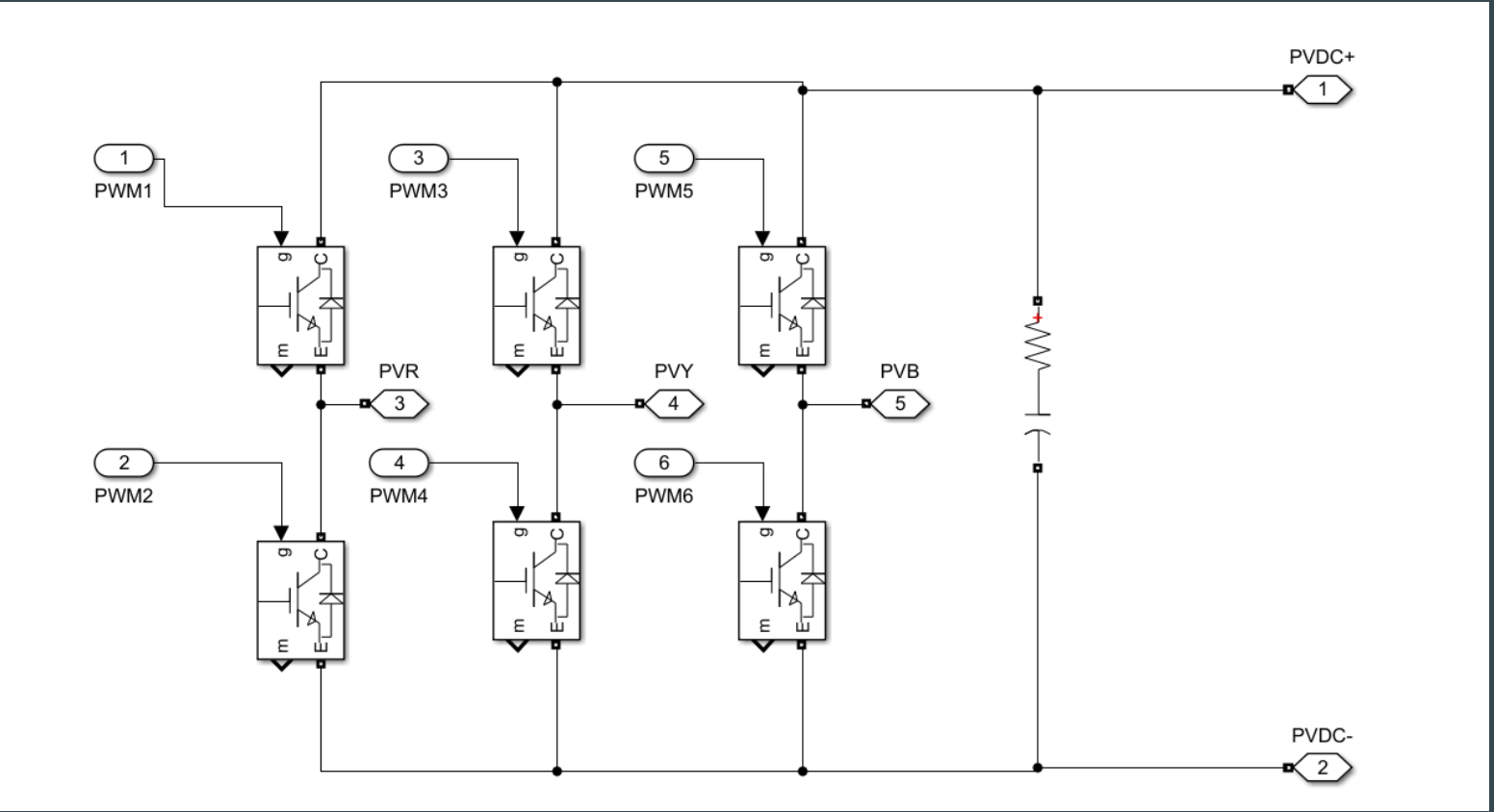

q-axis current controller1
0
Reactive Power=0 of Inv
error
PI(s)
q-axis current controller
[IQ]
PV Inv (Baxis) Actual current
Ref Q axis Vinductor
-K-
[ID]
Vref (q)
[VQ]
grid Voltage (q-ax)
2
[BVDC]
x
Vq(ref)

CONTROLLER TO PWM GENERATION



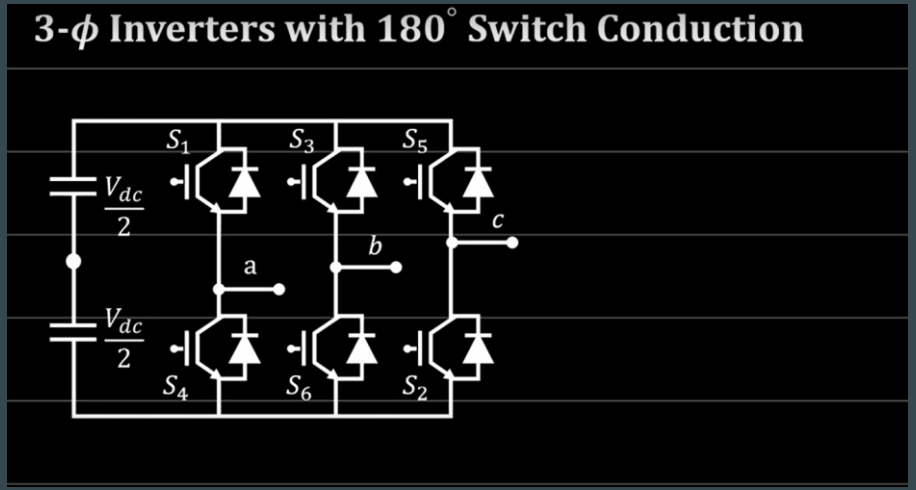


Inside Block Diagram



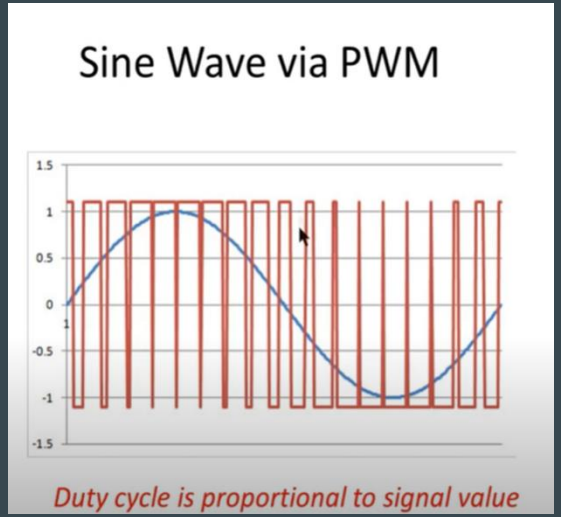
Circuit Information

- There 6 switching sequence.
- There are six modes of operation in a cycle and the duration of each mode is 60°.
- Switches of any leg can't be switch on same time.



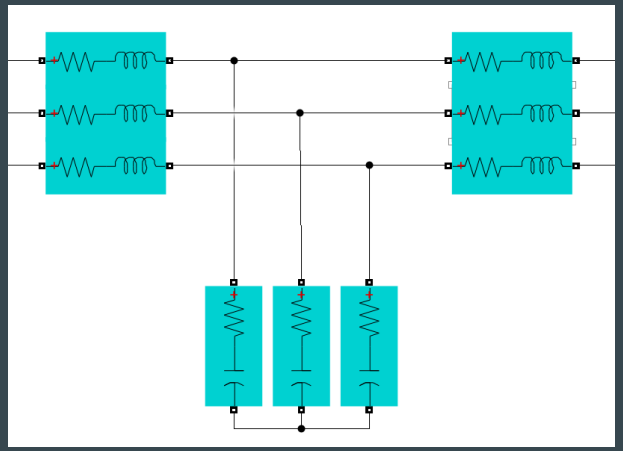
Working Principle

- The Input for the switching circuit is PWM signal.
- The PWM signal is generated according to the d axis and q axis current.
- The IGBT switches are being controlled by the PWM signals generated at the PI control.
- The duty cycle is proportional to signal value.

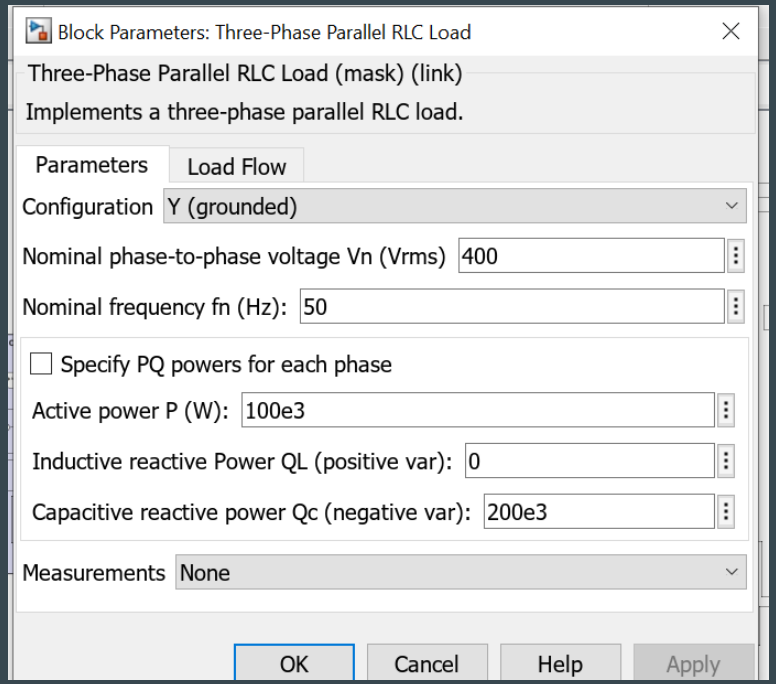


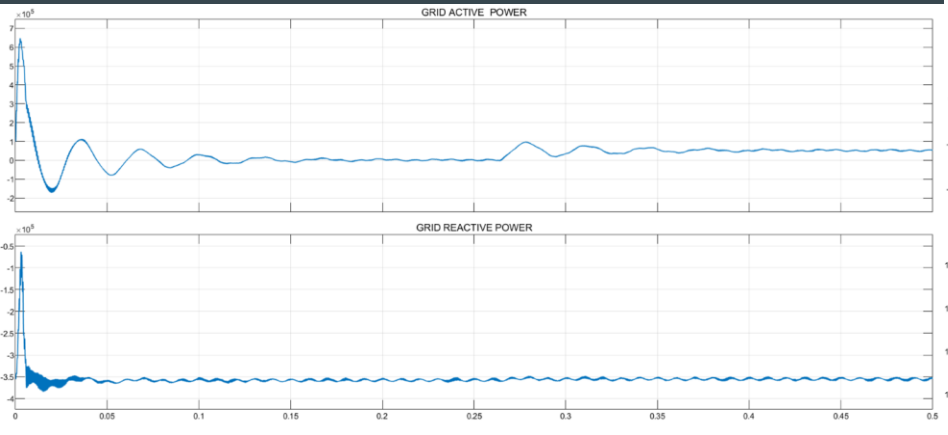
Harmonic Filter

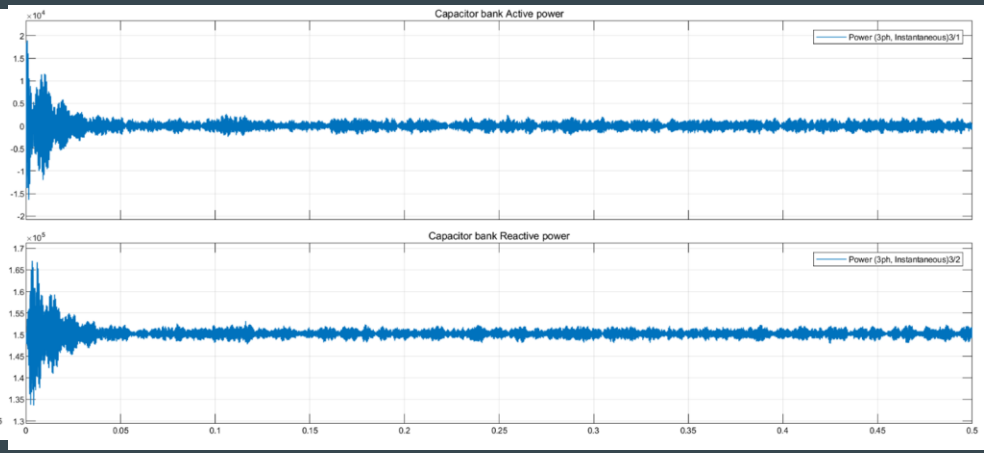
- Input side Inductor (L1) limits the rate of change of the signal reducing high-frequency components..
- Capacitor provides a low-impedance path for high-frequency currents so that harmonics flow through the filter rather than reaching the load.
- Output side Inductor (L2) attenuates high-frequency components and provides a return path for these frequencies back to the source.

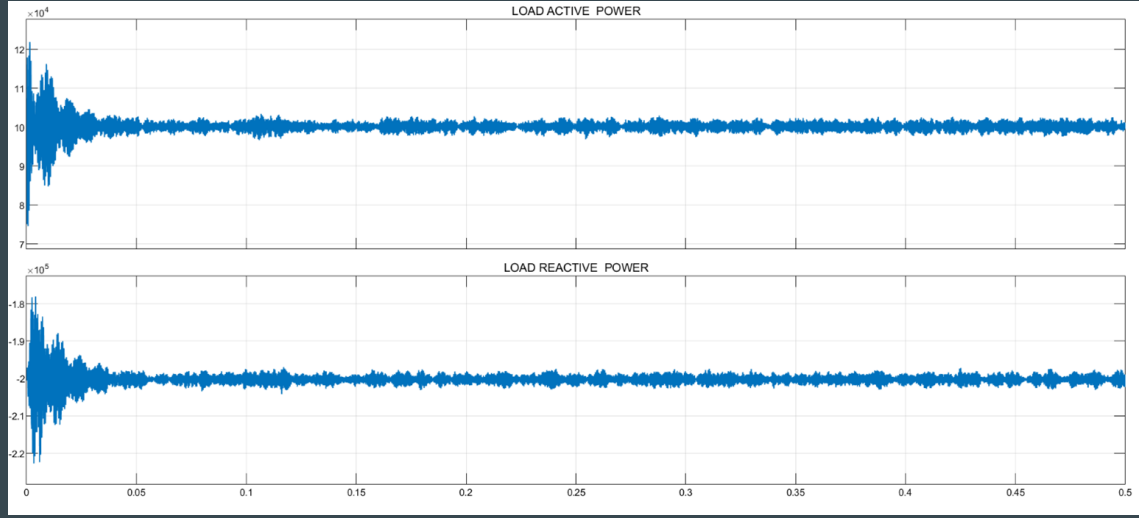


Capacitor Bank









Capacitive Load: 200k

Capacitor bank supplies: 150k

Grid supplies rest: 50k

STATCOM

It is build based on power electronics Voltage Source Converter and can act as either a source or sink of reactive AC power which is tied to a transmission line.

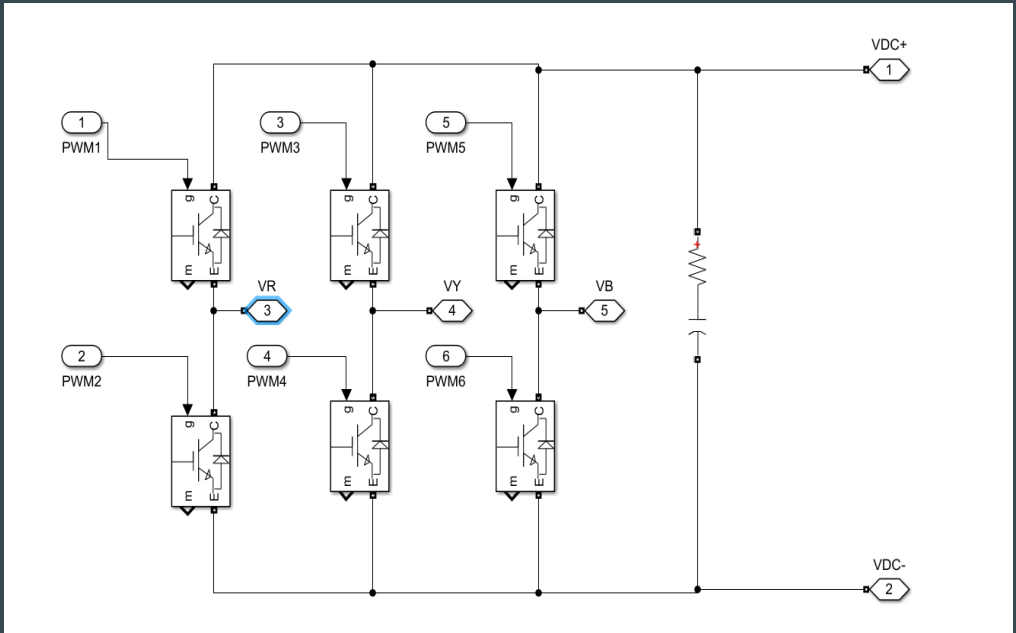
The STATCOM regulates the voltage magnitude at its terminals by controlling the amount of reactive power injected into or absorbed from the power system. When system voltage is low, the STATCOM generates reactive power

The charged capacitor provides a dc voltage to the converter, which produces a set of controllable three phase output voltages with frequency of the ac power system.

By varying the amplitude of the output voltage , the reactive power exchange between the converter and the AC system can be controlled. If the amplitude of the output voltage is increased above that of the AC system ,a leading current is produced.

Statcom inverter

In our model we designed, 3 phase 3 wire vsc inverter



Design parameter

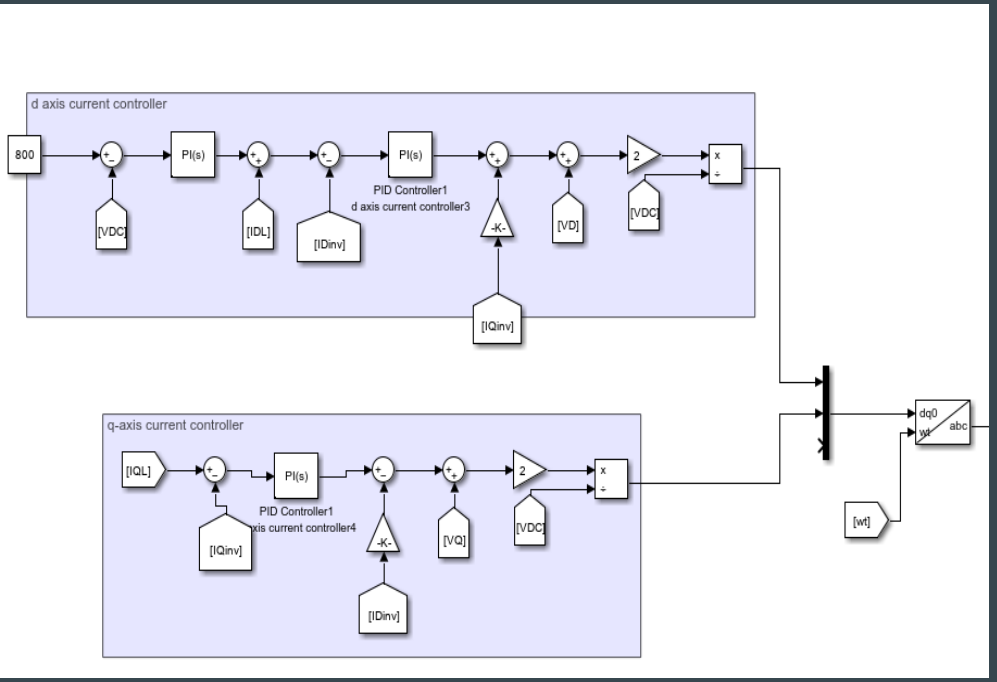
Vdc>= 2*(√2/3) Vl/m ; where m is the modulation index

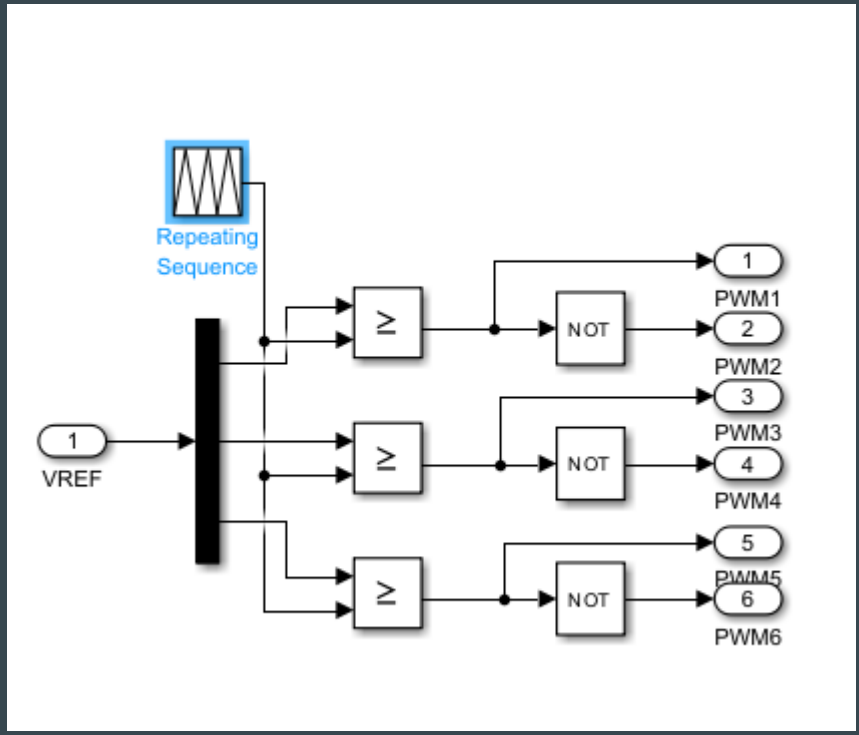
Assuming 10 percent loss in igbt switches,vdc ref is taken as 800v

This dc link voltage is then converted to 3 phase ac through pwm inverter

Switching frequency of carrier signal in pwm inverter is taken 10khz

Controller


d axis current controller
800
PI(s)
PID Controller1
d axis current controller3
[VDC]
[IDL]
[IDinv]
[IQinv]
[VD]
[VDC]
2
x
÷
q-axis current controller
[IQL]
PI(s)
PID Controller1
[IQinv]
[IDinv]
[VQ]
[VDC]
2
[wt]
dq0
abc


Repeating
Sequence
1
VREF
≥
NOT
1
PWM1
2
PWM2
3
PWM3
4
PWM4
5
PWM5
6
PWM6

Results:

Will be shown during demonstration

references

chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/http://pe.org.pl/articles/2012/12a/50.pdf

chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://benthamopen.com/contents/pdf/TOAUTOCJ/TOAUTOCJ-6-1415.pdf